

Answer Key & Teacher Notes

Session 1 – Comparing Ratios with Common Terms • Session 2 – Comparing Rates with Benchmarks

HOW TO USE THIS LESSON'S PAGES

one lesson = two sessions

Warm-Up	First 5–8 minutes of Session 1. Key: Unit Warm-Up Key.
Prerequisite Practice A/B	Before the lesson, for students who need the skill the session uses. Key in that folder.
Guided Notes	During the lesson. Students fill in the blanks as you teach; answers below.
Practice Form A	In class, after the lesson. Two pages. The self-check code tells students if they are on track.
Practice Form B	Same skills, new numbers: reteach, retake, small group or absent students.
Exit Ticket	Last 5 minutes. Two per page; cut in half.
Homework Packet	Sheet 1 after Session 1, Sheet 2 after Session 2. Level 1 has the same problems and answers, with a word bank, step frames and larger type for students who need language support.

SESSION 1 • TEACHER NOTES

Comparing Ratios with Common Terms

TALK PROMPTS

- Why does making the red paint equal (6 cups) make the comparison so simple?
- Both plant-food bottles use more water than plant food. Why does matching the drops settle the question?

WATCH FOR

- Comparing second terms without matching the first terms (" $7 > 5$, so B is greener").
- Adding to match the first terms (turning $2 : 5$ into $6 : 9$) instead of multiplying.
- Thinking the mix with more water is the stronger mix.

PARTNER TASK

Orange Juice, Bubble Solution, Plant Food. One partner chooses the common multiple and scales the first mix; the other scales the second mix. Together, compare the other term and say which mix is stronger.

SESSION 2 • TEACHER NOTES

Comparing Rates with Benchmarks

TALK PROMPTS

- Why is scaling to 30 tokens easier than dividing 7 by 3?
- Do the unit prices and the 12-bag totals point to the same store? Why should they agree?

WATCH FOR

- Subtracting the two quantities ($9 - 2$) instead of comparing rates.
- Choosing a benchmark that one amount cannot reach exactly (10 for a 4-pack and a 6-pack).
- Picking the larger total cost as the better buy instead of the lower cost for the same amount.

PARTNER TASK

Gummy Bears, Typing Contest, Corner Store Chips. One partner compares with unit rates; the other compares with a benchmark. Check that both methods name the same winner, then switch methods.

ITEM ANALYSIS

depth of knowledge (DOK) for each practice item

Sheet	Standard	1	2	3	4	5	6	7	8	9	10	11
S1 • Form A	6.RP.A.3b	1	1	2	2	2	2	2	3	3	1	2
S1 • Form B	6.RP.A.3b	1	1	2	2	2	2	2	3	3	1	1
S2 • Form A	6.RP.A.3b	1	1	2	2	2	2	2	3	3	1	1
S2 • Form B	6.RP.A.3b	1	1	2	2	2	2	2	3	3	1	1

Items before the last three are Practice; then Find the Error (DOK 3) and two Test Practice items.

GUIDED NOTES

common multiple: A number both terms can be **scaled** to.

least common multiple: The **smallest** number that is a multiple of both numbers.

stronger: More of the main ingredient for the **same** amount of water.

- To compare two ratios, scale them so one term is the **same** in both.
- Then compare the **other** term. Less water for the same soap means a **stronger** mix.

Example: A common multiple of 2 and 3 is **6**. Mix A $\times 3 = 6 : 15$. Mix B $\times 2 = 6 : 14$. Compare the water: 14 cups is less than 15 cups. Answer: Mix **B** is stronger: less water for the same 6 spoons of soap.

YOU TRY

- A: 6 : 27; B: 6 : 24. Tea B is stronger (less water).
- a. 35 b. 24

EXIT TICKET

- a. 15 b. 22
- A: 6 : 9; B: 6 : 10. Mix A (less water)
- B** (less water)

PRACTICE FORM A

- a. 32 b. 30 c. 27 d. 14
- Mix A $\times 2 = 6 : 8$; Mix B $\times 3 = 6 : 9$. Mix B is bluer (more blue for 6 cups red).
- a. Pitcher 1: 6 : 10; Pitcher 2: 6 : 9 b. Pitcher 2 (less water)
- a. M: 20 : 36; N: 20 : 35 b. Bottle N (less water for 20 drops)
- A: 6 : 18; B: 6 : 16. Cup B (less milk)
- Pool 2: 6 tablets for 10,000 gallons, more than Pool 1's 5
- Jar 1: 15 : 35; Jar 2: 15 : 36. Jar 1 is stronger (less water).
- a. Team 1: 20 : 45; Team 2: 20 : 48 b. Team 1, because it needs fewer games (45) for the same 20 wins.
- The white amounts are different, so 7 and 5 cannot be compared directly. At 6 cups white: A = 6 : 15, B = 6 : 14. Mix A is greener.
- B** (one term equal)
- A** (2 to 5)

Self-check word COMMON

PRACTICE FORM B

- a. 24 b. 63 c. 15 d. 20
- Mix A $\times 3 = 12 : 15$; Mix B $\times 4 = 12 : 16$. Mix B is bluer (more blue for 12 cups red).
- a. Pitcher 1: 12 : 21; Pitcher 2: 12 : 20 b. Pitcher 2 (less water)
- a. M: 15 : 40; N: 15 : 39 b. Bottle N (less water for 15 drops)
- A: 6 : 20; B: 6 : 21. Cup A (less milk)
- Pool 1: Pool 2 uses 6 tablets for 10,000 gallons, fewer than Pool 1's 7
- Jar 1: 6 : 15; Jar 2: 6 : 16. Jar 1 is stronger (less water).
- a. Team 1: 12 : 32; Team 2: 12 : 30 b. Team 2, because it needs fewer games (30) for the same 12 wins.
- The red amounts are different, so 10 and 8 cannot be compared directly. At 12 cups red: A = 12 : 32, B = 12 : 30. Mix A is bluer.
- B** (12 : 32)
- B** (less milk)

Self-check word EQUALS

GUIDED NOTES

benchmark: A shared amount you scale **both** choices to.

rate: Compares two amounts with **different** units.

unit rate: The amount for exactly **one**.

- Pick a benchmark that both amounts **divide** into evenly, like 20 for 2 and 4.
- Scale each rate to the benchmark, then **compare** the results.

Example: Benchmark: **20** minutes for both hoses. Hose A: $6 \times 10 = 60$ gallons. Hose B: $10 \times 5 = 50$ gallons. Compare: 60 gallons > 50 gallons in the same 20 minutes. Answer: Hose **A** fills faster.

YOU TRY

1. A: $\$8 \times 4 = \32 ; B: $\$10 \times 3 = \30 . Store B
2. a. 12 (any common multiple) b. 40 (any common multiple)

EXIT TICKET

1. a. 15 miles b. \$2.40
2. A: $\$3 \times 3 = \9 ; B: $\$5 \times 2 = \10 . Pack A
3. **A** (divide)

PRACTICE FORM A

1. a. 16 flyers b. 14 stairs c. \$2.50 d. 30 miles
2. A: $4 \times 15 = 60$ tickets; B: $7 \times 10 = 70$ tickets. Game B
3. Tyler: $140 \div 4 = 35$; Sarah: $175 \div 5 = 35$. They type at the same rate.
4. A: $\$15 \times 4 = \60 ; B: $\$18 \times 3 = \54 . Store B
5. a. J: \$2.25; K: \$2.20 b. J: \$27.00; K: \$26.40 c. Store K
6. A: $\$42.50 \div 5 = \8.50 ; B: $\$66.40 \div 8 = \8.30 . Rink B
7. a. Reader 1: $40 \times 6 = 240$; Reader 2: $54 \times 5 = 270$ b. Reader 2
8. a. A: $\$120 \times 11 = \$1,320$; B: $\$176 \times 8 = \$1,408$ b. Company A (\$15 vs. \$16 per student) c. Sample: it keeps whole numbers, so there is no long division or rounding.
9. Subtracting does not compare rates. Use a 10-ticket benchmark: A = $9 \times 5 = 45$ points; B = $15 \times 2 = 30$ points. Ride A gives more points per ticket (4.5 vs. 3).
10. **B** (12)
11. **B** (Store Y)

Self-check word COMPARE

PRACTICE FORM B

1. a. 15 flyers b. 12 stairs c. \$2.25 d. 32 miles
2. A: $5 \times 15 = 75$ stickers; B: $8 \times 10 = 80$ stickers. Machine B
3. Tyler: $156 \div 4 = 39$; Sarah: $190 \div 5 = 38$. Tyler
4. A: $\$9 \times 3 = \27 ; B: $\$14 \times 2 = \28 . Store A
5. a. J: \$2.50; K: \$2.40 b. J: \$30.00; K: \$28.80 c. Store K
6. A: $\$34.00 \div 4 = \8.50 ; B: $\$52.20 \div 6 = \8.70 . Rink A
7. a. Reader 1: $35 \times 6 = 210$; Reader 2: $48 \times 5 = 240$ b. Reader 2
8. a. A: $\$120 \times 11 = \$1,320$; B: $\$165 \times 8 = \$1,320$ b. Neither; both cost \$15 per student. c. Sample: it keeps whole numbers, so there is no long division or rounding.
9. Subtracting does not compare rates. Use a 15-ticket benchmark: A = $12 \times 5 = 60$ points; B = $15 \times 3 = 45$ points. Ride A gives more points per ticket (4 vs. 3).
10. **C** (15)
11. **A** (reach exactly)

Self-check word BARGAIN

HOMEWORK PACKET • ANSWERS

same answers for the core packet and Level 1

SHEET 1 · COMPARING RATIOS WITH COMMON TERMS

1. a. 40 b. 42 c. 44 d. 27
2. P yellow: 8, 12, 16, 20; Q yellow: 12, 18, 24, 30. At 15 cups blue, P has 20 yellow and Q has 18 yellow, so Mix Q is bluer.
3. a. $9 : 12 \times 4 = 36 : 48$ b. $13 : 16 \times 3 = 39 : 48$ c. Pack B (39 of 48 sprout)
4. No. Scale to 12 spoons sugar: $X = 12 : 27$, $Y = 12 : 30$. X has less water for the same sugar, so Drink X is sweeter.

SHEET 2 · COMPARING RATES WITH BENCHMARKS

1. a. 12 b. 18 (any common multiple is correct)
2. Kai: $90 \times 3 = 270$ words; Lena: $140 \times 2 = 280$ words. Lena types faster.
3. a. A: $\$8 \times 8 = \64 ; B: $\$12 \times 5 = \60 b. Store B
4. a. C: $\$7 \times 4 = \28 ; D: $\$9 \times 3 = \27 b. Store D
5. Benchmark 40 gallons: Car 1 goes $150 \times 8 = 1,200$ miles; Car 2 goes $200 \times 5 = 1,000$ miles. Car 1 is more efficient.
6. **B** (24 bagels)
7. **B** (Ella (180 min))